Examiner only

4. A Welsh farmer decides to purchase a wind generator to provide some electrical energy for his farm.
A selection of wind generators is available.

	Wind generator 1	Wind generator 2	Wind generator 3
Cost to buy and install (£)	1800	3000	2925
Expected lifetime (years)	15	15	15
Estimated number of units produced per year (kWh)	2000	4000	3000
Cost per unit (£)	0.15	0.15	0.15
Estimated saving per year (£)		600	450
Estimated payback time (years)		5.0	6.5

(a) (i) Use the equation:

estimated saving per year = units produced per year × cost per unit

and data from the table to calculate the estimated saving per year if **wind generator 1** was installed. [1]

Estimated saving per year = £ ....

(ii) Use the equation:

$$\text{estimated payback time} = \frac{\text{cost to buy and install}}{\text{estimated saving per year}}$$

and data from the table to calculate the estimated payback time for **wind generator 1**. [2]

Estimated payback time = ... years